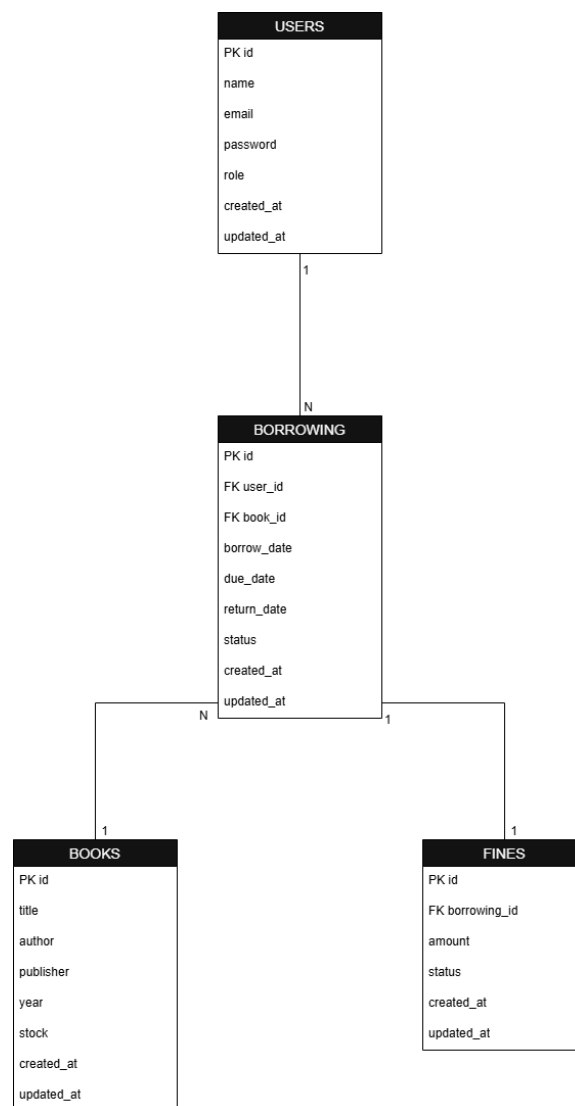


DOKUMENTASI BACKEND SISTEM PERPUSTAKAAN

A. Deskripsi Sistem

Sistem ini merupakan sistem backend berbasis REST API yang digunakan untuk mengelola proses peminjaman buku. Proyek ini dikembangkan menggunakan framework Laravel 13. Sistem ini menerapkan konsep Authentication menggunakan Laravel Sanctum dan Role-Based Access Control (RBAC) untuk membedakan hak akses Admin dan User.

B. ERD Database

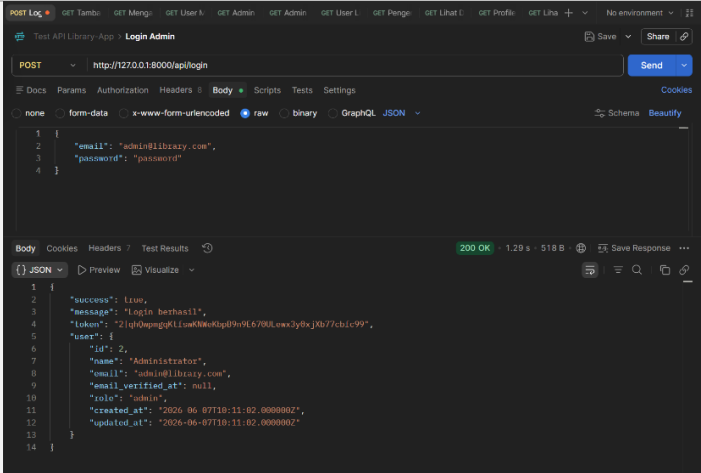
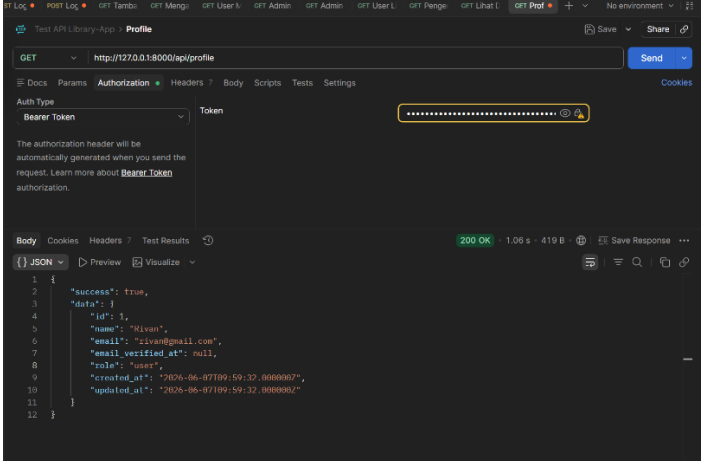
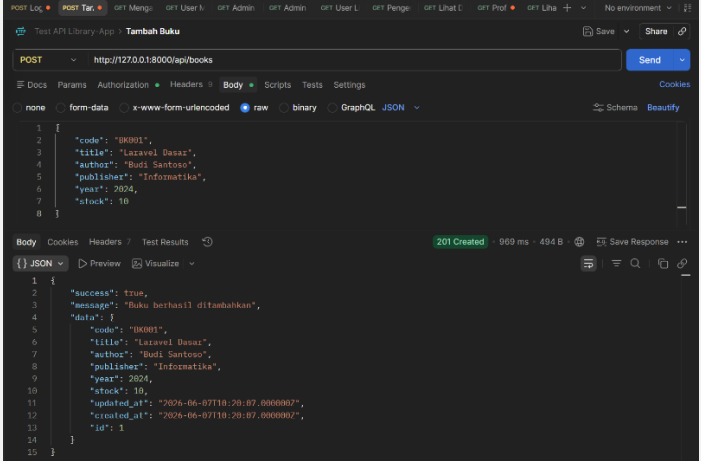


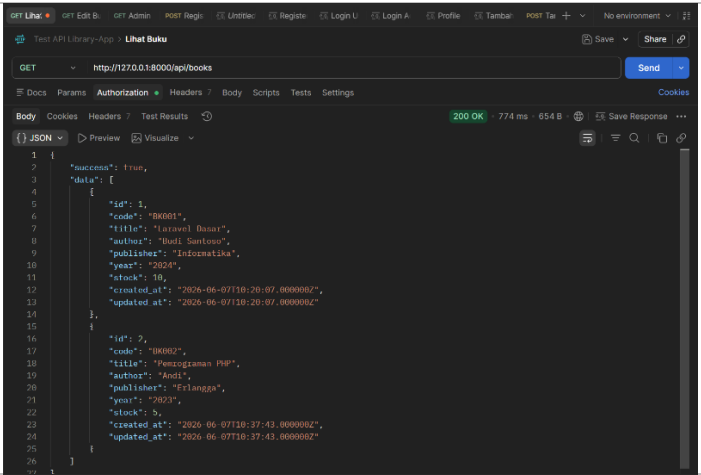
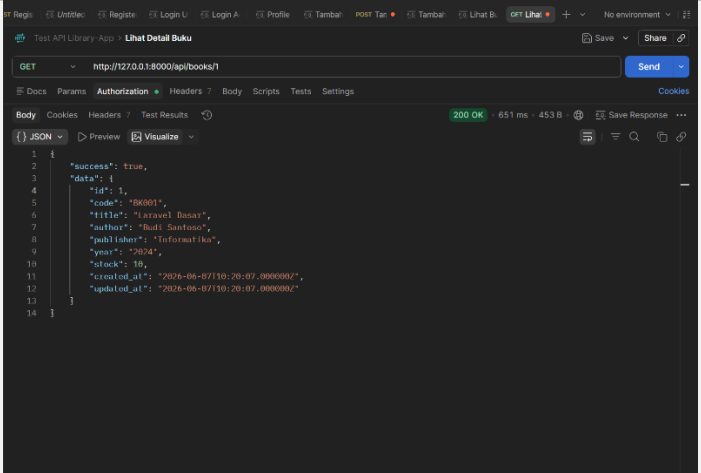
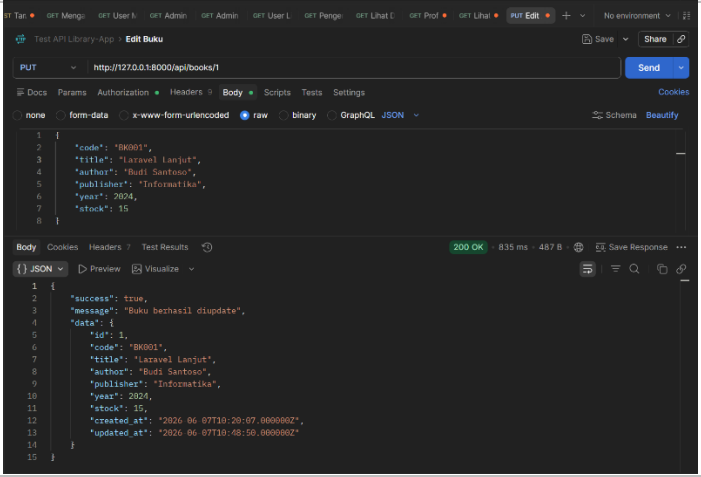
Kardinalitas Relasi

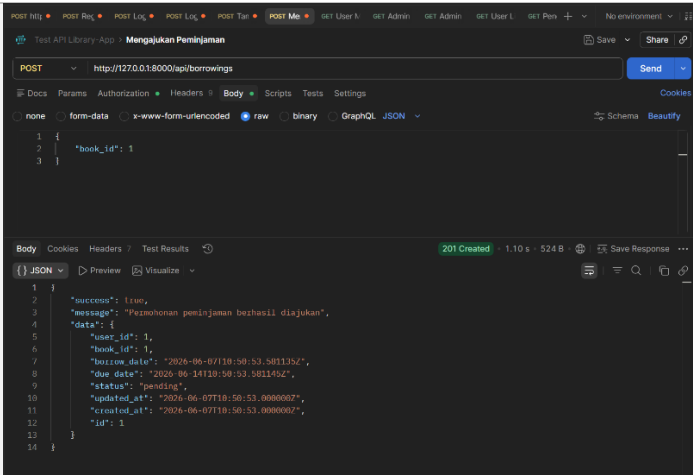
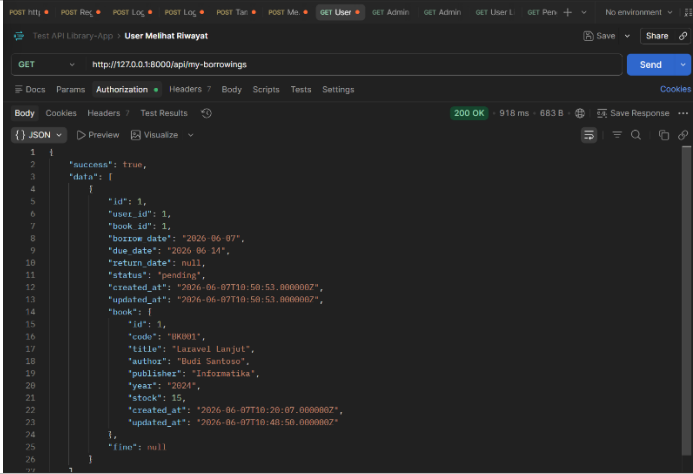
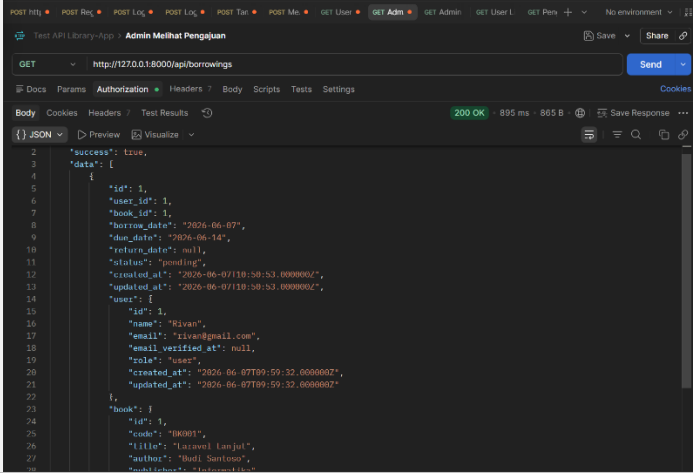
Tabel 1	Relasi	Tabel 2	Penjelasan
Users	1 : N	Borrowings	Satu anggota dapat melakukan banyak peminjaman
Books	1 : N	Borrowings	Satu buku dapat dipinjam berkali-kali
Borrowings	1 : 1	Fines	Satu transaksi peminjaman dapat memiliki satu denda

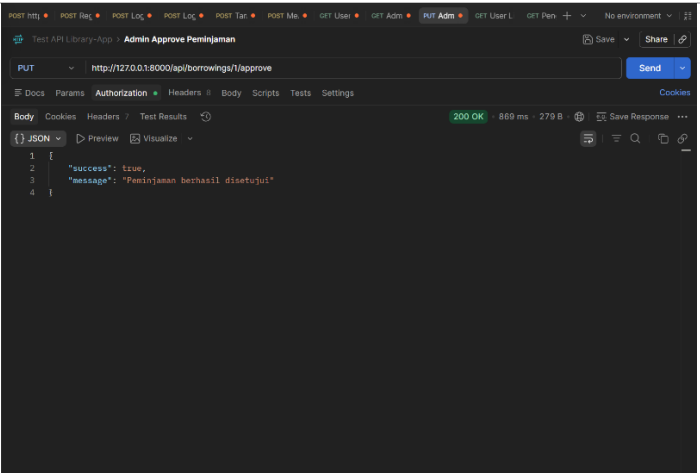
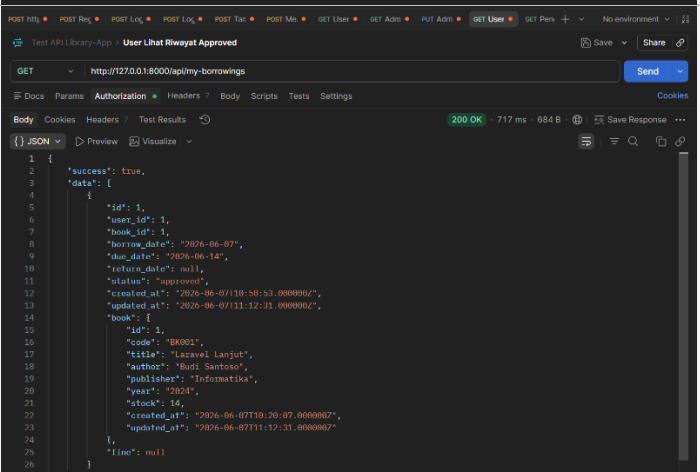
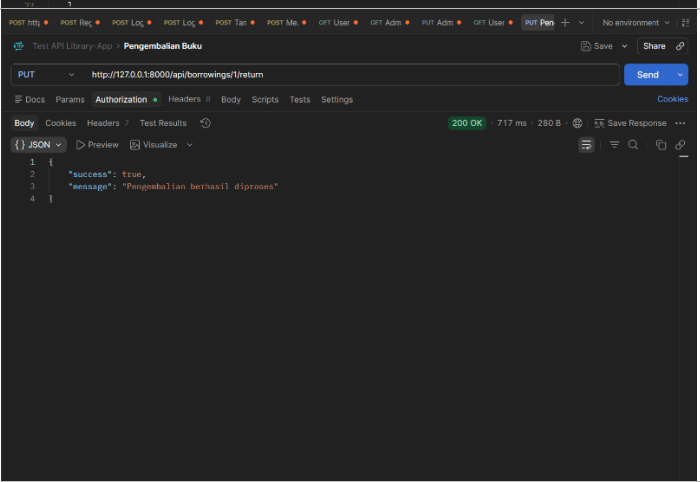
C. Endpoint API

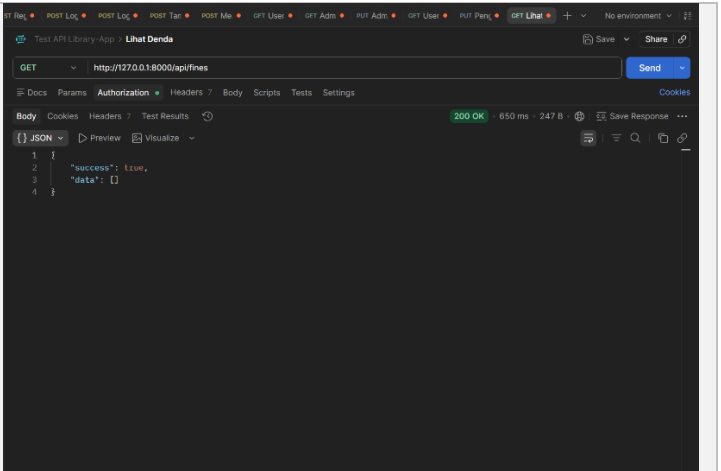
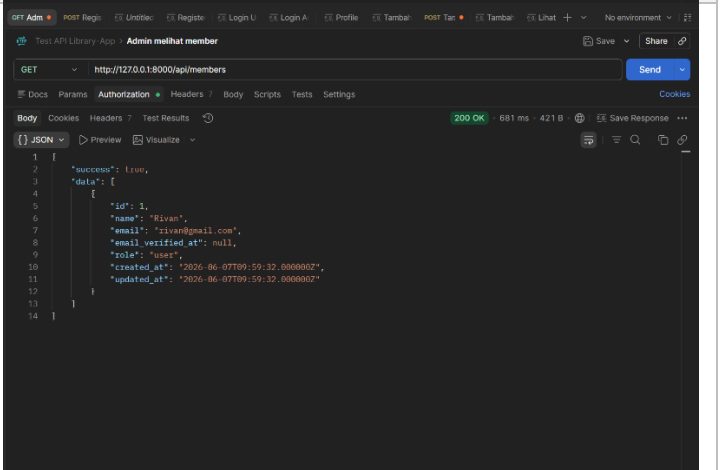
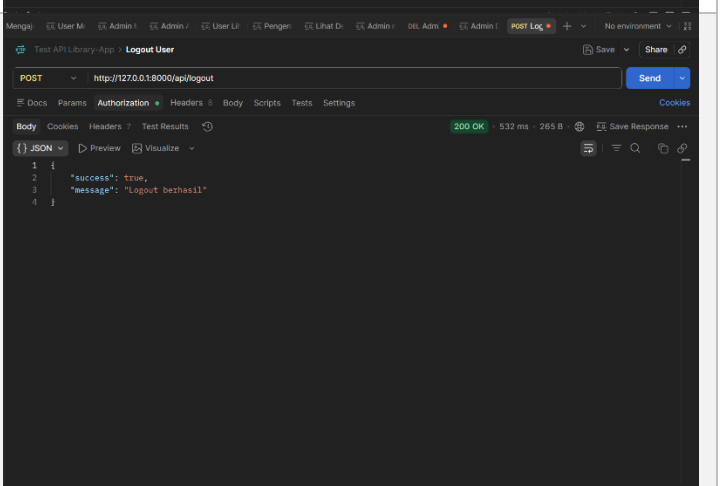
No	Method	Endpoint	Deskripsi
1	POST	/api/register	Mendaftarkan anggota perpustakaan baru ke dalam sistem.
2	POST	/api/login	Melakukan autentikasi pengguna dan menghasilkan token Sanctum.
3	POST	/api/logout	Menghapus token aktif sehingga pengguna keluar dari sistem.
4	GET	/api/profile	Menampilkan informasi pengguna yang sedang login.
5	GET	/api/books	Menampilkan seluruh data buku yang tersedia di perpustakaan.
6	POST	/api/books	Menambahkan data buku baru ke dalam sistem.
7	GET	/api/books/{id}	Menampilkan detail satu buku berdasarkan ID.
8	PUT	/api/books/{id}	Mengubah informasi buku yang sudah tersimpan.
9	DELETE	/api/books/{id}	Menghapus data buku dari sistem.
10	GET	/api/members	Menampilkan seluruh anggota perpustakaan yang terdaftar.
11	GET	/api/members/{id}	Menampilkan detail anggota berdasarkan ID.
12	DELETE	/api/members/{id}	Menghapus data anggota perpustakaan.
13	POST	/api/borrowings	Digunakan user untuk mengajukan peminjaman buku.
14	GET	/api/my-borrowings	Menampilkan seluruh riwayat peminjaman milik pengguna yang sedang login.

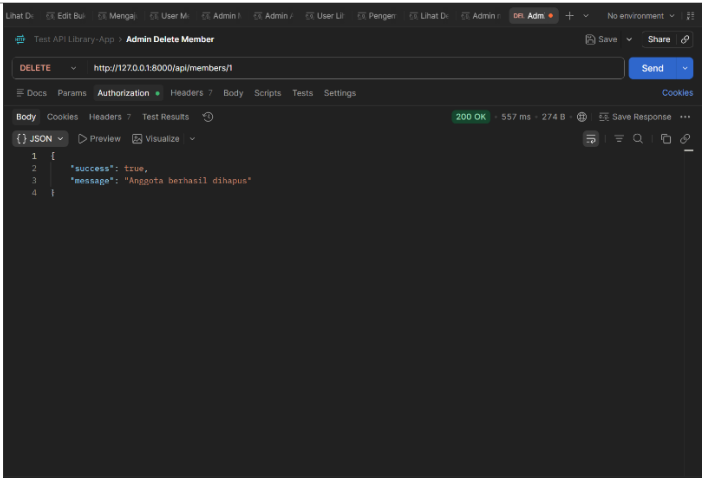
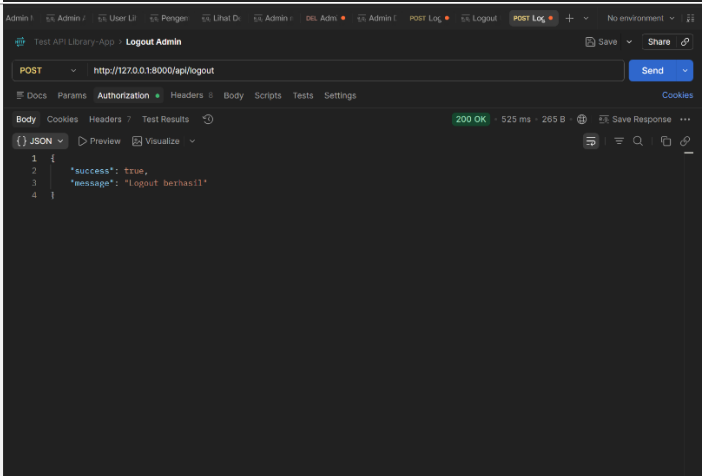
3	http://127.0.0.1:8000/api/login	 POST Log GET Tambu GET Mengi GET User Admin GET Admin GET User L GET Peng GET Lhat GET Lhat + No environment Test API Library App Login Admin POST http://127.0.0.1:8000/api/login Send Body raw <pre> 1 { 2 "email": "admin@library.com", 3 "password": "password" 4 } </pre> Body JSON Preview Visualize <pre> 1 { 2 "success": true, 3 "message": "login berhasil", 4 "token": "Zlg0wpmgkLlswKkxk9b9rRl6700Lew3y0xjX077cbic99", 5 "user": { 6 "id": 2, 7 "name": "Administrator", 8 "email": "admin@library.com", 9 "email_verified_at": null, 10 "role": "admin", 11 "created_at": "2026-06-07T10:11:02.000000Z", 12 "updated_at": "2026-06-07T10:11:02.000000Z" 13 } 14 } </pre>
4	http://127.0.0.1:8000/api/profile	 GET Log POST Log GET Tambu GET Mengi GET User Admin GET Admin GET User L GET Peng GET Lhat GET Lhat + No environment Test API Library App Profile GET http://127.0.0.1:8000/api/profile Send Auth Type Bearer Token Token The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization. Body JSON Preview Visualize <pre> 1 { 2 "success": true, 3 "data": { 4 "id": 1, 5 "name": "Rivan", 6 "email": "rivan@gmail.com", 7 "email_verified_at": null, 8 "role": "user", 9 "created_at": "2026-06-07T09:59:32.000000Z", 10 "updated_at": "2026-06-07T09:59:32.000000Z" 11 } 12 } </pre>
5	http://127.0.0.1:8000/api/books	 POST Log POST Log GET Tambu GET Mengi GET User Admin GET Admin GET User L GET Peng GET Lhat GET Lhat + No environment Test API Library App Tambah Buku POST http://127.0.0.1:8000/api/books Send Body raw <pre> 1 { 2 "code": "BK001", 3 "title": "Lazavel Besar", 4 "author": "Budi Santoso", 5 "publisher": "Informatika", 6 "year": 2024, 7 "stock": 10 8 } </pre> Body JSON Preview Visualize <pre> 1 { 2 "success": true, 3 "message": "Buku berhasil ditambahkan", 4 "data": { 5 "code": "BK001", 6 "title": "Lazavel Besar", 7 "author": "Budi Santoso", 8 "publisher": "Informatika", 9 "year": 2024, 10 "stock": 10, 11 "updated_at": "2026-06-07T10:20:07.000000Z", 12 "created_at": "2026-06-07T10:20:07.000000Z", 13 "id": 1 14 } 15 } </pre>

6	http://127.0.0.1:8000/api/books	
7	http://127.0.0.1:8000/api/books/1	
8	http://127.0.0.1:8000/api/books/1	

9	http://127.0.0.1:8000/api/borrowings	 <pre> 1 { 2 "success": true, 3 "message": "Peminjaman pinjaman berhasil diajukan", 4 "data": { 5 "user_id": 1, 6 "book_id": 1, 7 "borrow_date": "2026-06-07T18:58:53.581135Z", 8 "due_date": "2026-06-14T18:58:53.581135Z", 9 "status": "pending", 10 "updated_at": "2026-06-07T18:58:53.800000Z", 11 "created_at": "2026-06-07T18:58:53.800000Z", 12 "id": 1 13 } 14 } </pre>
10	http://127.0.0.1:8000/api/my-borrowings	 <pre> 1 { 2 "success": true, 3 "data": [4 { 5 "id": 1, 6 "user_id": 1, 7 "book_id": 1, 8 "borrow_date": "2026-06-07", 9 "due_date": "2026-06-14", 10 "return_date": null, 11 "status": "pending", 12 "created_at": "2026-06-07T18:58:53.800000Z", 13 "updated_at": "2026-06-07T18:58:53.800000Z", 14 "book": { 15 "id": 1, 16 "code": "BK0001", 17 "title": "Lazwadi Lanjut", 18 "author": "Budi Santoso", 19 "publisher": "Informatika", 20 "year": "2024", 21 "stock": 15, 22 "created_at": "2026-06-07T18:28:07.000000Z", 23 "updated_at": "2026-06-07T18:48:58.000000Z", 24 }, 25 "film": null 26 } 27] 28 } </pre>
11	http://127.0.0.1:8000/api/borrowings	 <pre> 2 "success": true, 3 "data": [4 { 5 "id": 1, 6 "user_id": 1, 7 "book_id": 1, 8 "borrow_date": "2026-06-07", 9 "due_date": "2026-06-14", 10 "return_date": null, 11 "status": "pending", 12 "created_at": "2026-06-07T18:58:53.800000Z", 13 "updated_at": "2026-06-07T18:58:53.800000Z", 14 "user": { 15 "id": 1, 16 "name": "Rivan", 17 "email": "rivan@gmail.com", 18 "email_verified_at": null, 19 "role": "user", 20 "created_at": "2026-06-07T09:59:32.000000Z", 21 "updated_at": "2026-06-07T09:59:32.000000Z" 22 }, 23 "book": { 24 "id": 1, 25 "code": "BK0001", 26 "title": "Lazwadi Lanjut", 27 "author": "Budi Santoso", 28 "publisher": "Informatika", 29 "year": "2024", 30 "stock": 15, 31 "created_at": "2026-06-07T18:28:07.000000Z", 32 "updated_at": "2026-06-07T18:48:58.000000Z", 33 } 34 } 35] 36 } </pre>

12	http://127.0.0.1:8000/api/borrowings/1/approve	
13	http://127.0.0.1:8000/api/my-borrowings	
14	http://127.0.0.1:8000/api/borrowings/1/return	

15	http://127.0.0.1:8000/api/fines	
16	http://127.0.0.1:8000/api/members	
17	http://127.0.0.1:8000/api/logout	

18	http://127.0.0.1:8000/api/members/2	 Test API Library-App - Admin Delete Member DELETE http://127.0.0.1:8000/api/members/2 Send Body Cookies Headers Test Results 200 OK - 557 ms - 274 B <pre>{ "success": true, "message": "Anggota berhasil dihapus" }</pre>
19	http://127.0.0.1:8000/api/logout	 Test API Library-App - Logout Admin POST http://127.0.0.1:8000/api/logout Send Body Cookies Headers Test Results 200 OK - 525 ms - 265 B <pre>{ "success": true, "message": "Logout berhasil" }</pre>